

SINA DILEK

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PROFILE

Second-year Computer Science student (First-Class Honours, top of cohort) currently interning as a Software Engineer at Cloudflare, and CTO of a pre-seed AI startup where I own architecture, backend infrastructure, and engineering team direction. I work across the full stack but my main interest is backend systems, data pipelines, and developer tooling. Comfortable building real-world AI-integrated products and working at the intersection of engineering and product thinking.

EDUCATION

University of Essex

BSc Computer Science, First-Class Honours (Expected 2027)

Colchester, United Kingdom

Feb 2024 – Present

- Top of cohort across two years; **Dean's List for Excellence** (year average 80).
- First-class marks across systems, algorithms, and web/software engineering modules, including Web Application Programming (94) and Application Programming (91).
- Relevant modules: Algorithms & Data Structures, Databases & Information Retrieval, Software Engineering, Computer Security, Introduction to AI.
- Completed Boot.dev Back-End Development Path covering Python, Go, Linux, and core computer science fundamentals.

EXPERIENCE

Software Engineer Intern

Cloudflare — Resilience (Foundational Engineering) Org

Jun 2026 – Present

London, United Kingdom

- Software engineer on a foundational engineering team, shipping to a production internal codebase in **TypeScript** on **Cloudflare Workers** and **D1**, with work spanning backend services, testing, and internal developer tooling.
- Contributed merged pull requests against live tickets within the first days, including expanding automated test coverage and fixing a silently-failing section of the end-to-end suite.
- Own a greenfield internal tooling project end-to-end — from design and coding standards through implementation — building static-analysis-based dependency mapping across TypeScript, Go, Rust, and Python codebases.

Chief Technology Officer (CTO)

Price Lantern — AI-Powered Quote Analysis Platform · Pre-seed Startup

Nov 2025 – Present

Remote

- Founding engineer turned CTO at a pre-seed B2B AI platform for the construction and trade industry. Own full technical direction, infrastructure, and a small engineering team — hired and manage one direct report via task delegation, PR reviews, and technical mentorship.
- Leading a full **v2 rewrite** in **FastAPI**, **Supabase**, and **React**, replacing the legacy **Django/Firebase** stack after identifying scalability bottlenecks in the original architecture.
- Designed a cache-first retail enrichment pipeline with parallel scraping across UK retailers (Screwfix, Toolstation, Wickes), Firebase TTL caching, and batched persistence — cold lookups run **~2x faster** than the old sequential approach, cached repeat queries **80–95% faster**. Running scraper and AI search in parallel cuts that phase by a further **~40–50%**.
- Built async document processing with background job execution and status tracking so users get an instant response instead of blocking on the full pipeline, and hardened the AI integration with retry logic, timeouts, and model upgrades for invoice/quote extraction from PDFs.
- Architected product matching with **16-category classification**, multi-algorithm fuzzy matching (rapidfuzz), and cross-supplier unit normalisation; processed **2,000+ real-world invoices and quotes** to improve extraction consistency on messy document formats.
- Built the test suite (**957+ lines**) covering normalisation, matching, and edge cases, plus **724 lines** of architecture and onboarding docs so future engineers aren't lost.

SELECTED PROJECTS

Tayf — Real-Time Turkish News Bias Analyser | [tayfhaber.com](#) | [GitHub](#)

2026

- Solo full-stack “Ground News for Turkey” — ingests RSS from **144 Turkish news outlets** across the full political spectrum into a continuously-updating Postgres store, with a fault-tolerant pipeline of **per-feed circuit breakers**, **ETag/If-Modified-Since conditional requests**, and **bounded-concurrency worker pools** so one dead source can’t stall the rest.
- Same-story clustering is the genuinely hard part: built a **3-method ensemble** combining **Turkish character n-gram fingerprinting**, **TF-IDF cosine similarity over a rolling 48-hour window**, and a **named-entity overlap heuristic** to group cross-outlet coverage of the same event — with **no paid embedding or ML APIs**.
- Engineered a bias engine mapping outlets across **10 categories rolled up to 3 political-spectrum zones**, plus **blindspot detection** (events one side underreports) and a **cross-spectrum detector** that flags when an outlet breaks from its usual editorial line.
- Shipped a bilingual (TR/EN) **Next.js 16 + React 19** app: 10 pages, 7 API routes, password-gated admin dashboard, and health/metrics endpoints — backed by a **24-migration versioned Postgres schema** on Supabase.
- **GitHub Actions CI with CodeQL security scanning** and **118 automated tests across 8 Vitest suites** covering ingestion, clustering, bias mapping, and edge cases. Built solo in ~3.5 weeks.

Toka — AI Agent Harness with a JavaScript REPL Tool Surface | [GitHub](#)

2026

- Agent harness built on a contrarian premise: most harnesses give the model 20 narrow tools and the model contorts tasks to fit them. Toka exposes **one MCP tool** — a curated JavaScript REPL — and lets the model write, version, test, and discard its own tools as it works. The harness stays small; the space of what the agent can do inside it doesn’t.
- Implemented a **Node vm-based sandbox** for executing model-authored JS, append-only audit logging of every `exec` call, **pattern-based safety guardrails**, and a human approval gate for destructive operations — safe enough to let an LLM run arbitrary JS against a real working directory.
- Designed a **cross-session memory layer** so the agent improves over time: structured notes with file/line refs, tags, and `staleAfter` TTLs; a **personal tool registry with versioning** (last 10 versions per tool) and unit-test cases; a `per-persona self.md` the agent maintains itself with version history.
- Shipped on top of the **Claude Agent SDK** as a TypeScript CLI with an **Ink-based TUI** (slash commands, session resume, approval modals), headless mode with JSON output, and direct-API fallbacks — runs on the user’s existing Claude Code subscription, no separate key required.
- Project doubles as a design-philosophy artefact (`PRINCIPLES.md`, load-bearing system prompt) where the long-term roadmap is generalising the harness’s engine into a multi-provider protocol adapter for a downstream terminal product.

Lumiko — Auto-Documentation CLI | [GitHub](#)

2026

- Published CLI tool under **Mirako Studio** that auto-generates structured documentation (README, architecture, API reference) from any codebase by routing source through Claude — no AST parsing, no language-specific tooling.
- Produces a **machine-readable context.json** mapping every file’s purpose, exports, and function signatures so AI agents can ingest entire codebases without reading every file.
- Built in TypeScript with dual backends (**Claude Code CLI** by default for zero-API-cost runs, Anthropic API fallback), YAML config, and `tsup` for distribution as a global npm package.

Moji — Workspace-Centric Productivity App | [usemoji.app](#)

2025

- Full-stack productivity platform built with **FastAPI, Next.js, and Supabase (PostgreSQL)** — workspaces, tasks, notes, and rich-text pages.
- Secure auth with **JWT** and row-level security; modular REST API with typed schemas and a responsive TypeScript frontend with real-time updates.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, Java, C, SQL

Frameworks & Libraries: FastAPI, Django, React, Next.js, Node.js, Supabase, PostgreSQL, Firebase, Tailwind, rapidfuzz

Platforms & Tooling: Cloudflare Workers, D1, Git, GitHub Actions CI, LLM integration, MCP servers, Claude Agent SDK, document parsing pipelines